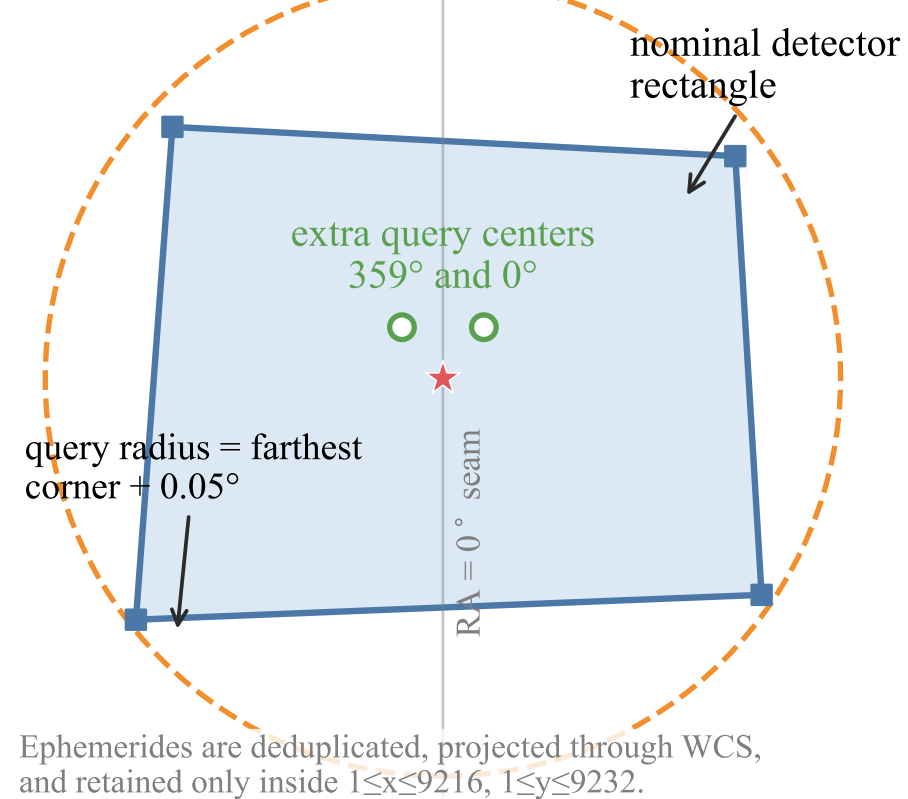
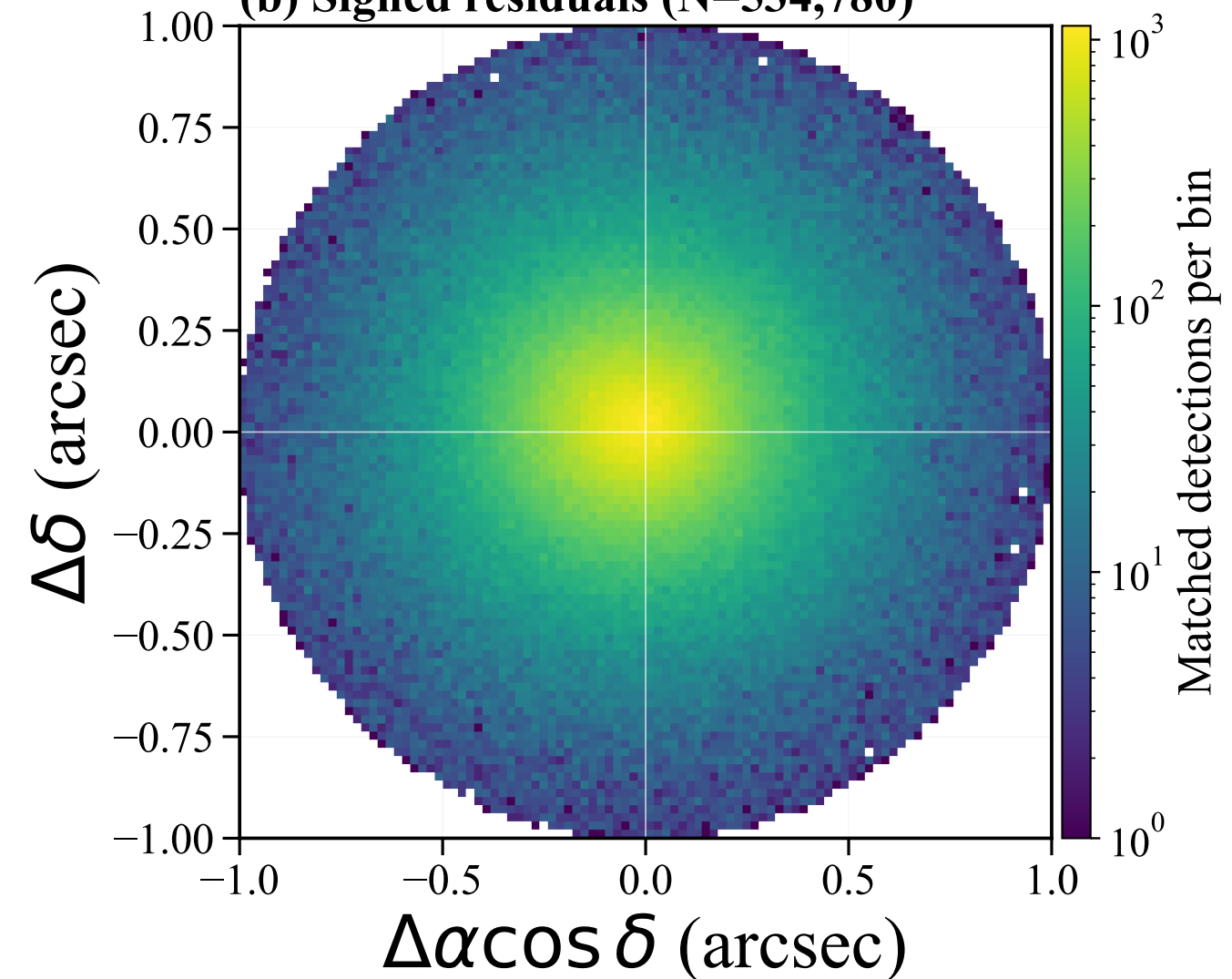


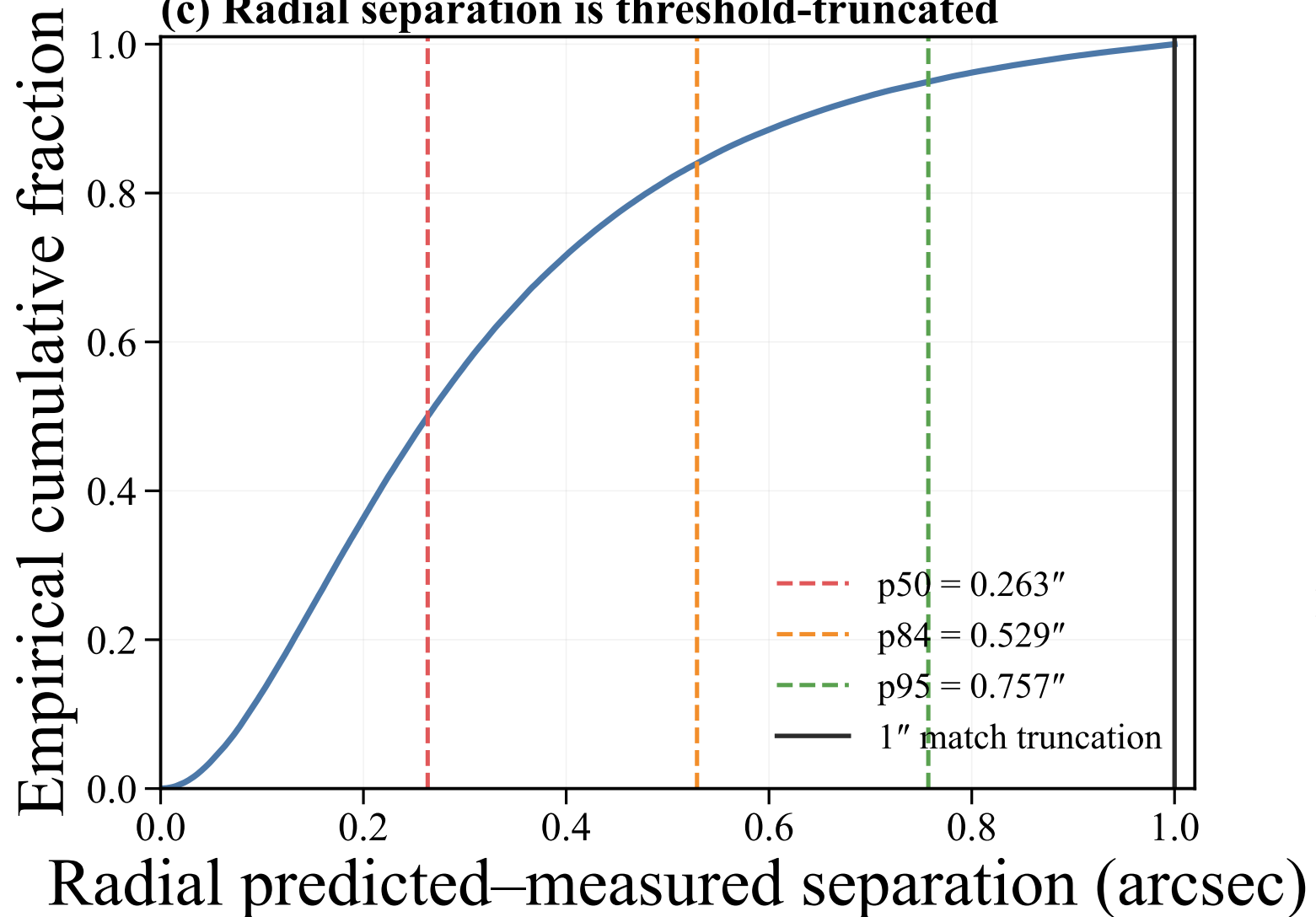
(a) Production query geometry (schematic)



(b) Signed residuals (N=534,780)



(c) Radial separation is threshold-truncated



(d) Position-shift chance-match control

